La Scuola d'Italia / Dr. Huson / IB Math 12 HL
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5.3 Quiz: Maclaurin Series and Kinematics

Answer all five problems. A calculator is allowed. Use the spaces provided for Problems 1 and 2. Answer Problems 3–5 on lined paper.

1. [Maximum mark: 6]

   Let $f(x) = (1 + 3x)\, e^{-2x}$.

   (a) Using the Maclaurin series for e^x, write down the first four terms of the Maclaurin series for e^{-2x}. [2]

   (b) Hence find the Maclaurin series for $f(x)$, up to and including the term in x^3. [3]

   (c) Hence write down the value of $f''(0)$. [1]

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2. [Maximum mark: 7]

A particle moves in a straight line. Its velocity, v m s^{-1}, at time t seconds is given by

$$v(t) = t^2 - 7t + 10, \qquad 0 \leq t \leq 6.$$

(a) Find the acceleration of the particle when $t = 4$. [2]

(b) Find the values of t for which the particle is at rest. [2]

(c) Find the total distance travelled by the particle for $0 \leq t \leq 6$. [3]

Nickname:

Answer Problems 3–5 on lined paper.

3. [Maximum mark: 13]

Let $f(x) = \ln(1 + \sin x)$, for $-\frac{\pi}{2} < x < \frac{3\pi}{2}$.

(a) Show that $f'(x) = \dfrac{\cos x}{1 + \sin x}$ and $f''(x) = -\dfrac{1}{1 + \sin x}$. [3]

(b) Find $f'''(x)$. [2]

(c) Hence find the Maclaurin series for $f(x)$, up to and including the term in x^3. [3]

(d) Use your answer to part (c) to find $\lim_{x \to 0} \dfrac{\ln(1 + \sin x) - x}{x^2}$. [2]

(e) Use your answer to part (c) to find an approximate value of $\int_0^{0.4} \ln(1 + \sin x)\, \mathrm{d}x$, and find the percentage error in this approximation. [3]

4. [Maximum mark: 14]

A car's engine is switched off at time $t = 0$, when its velocity is 30 m s^{-1}. For $t \geq 0$ its acceleration is $a = -0.02v^2$, where v m s^{-1} is its velocity at time t seconds.

(a) By solving a differential equation, show that $v = \dfrac{30}{1 + 0.6t}$. [4]

(b) Find the time at which the speed of the car is 12 m s^{-1}. [2]

(c) Find the distance travelled by the car in this time. [3]

(d) Given that $a = v\dfrac{\mathrm{d}v}{\mathrm{d}s}$, where s metres is the distance travelled since $t = 0$, find an expression for v in terms of s. [3]

(e) Hence find the speed of the car when it has travelled 100 metres. [2]

5. [Maximum mark: 13]

The function $y = f(x)$ satisfies the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x + 2y$, and $y = 1$ when $x = 0$.

(a) Use the differential equation to find the values of $f'(0)$, $f''(0)$ and $f'''(0)$. [4]

(b) Hence find the Maclaurin series for y, up to and including the term in x^3. [2]

(c) By using an integrating factor, solve the differential equation. Give your answer in the form $y = f(x)$. [5]

(d) Use your series from part (b) to estimate $f(0.2)$, and find the percentage error in this estimate. [2]